

## METER BRIDGE (01)

### Aims

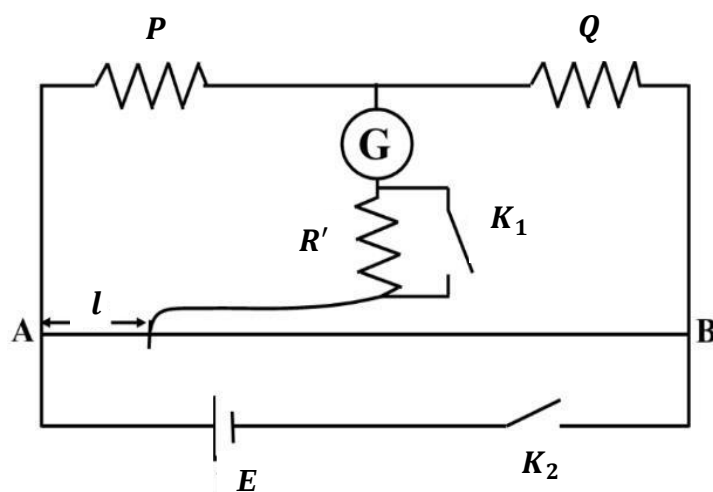
- A. To determine the end corrections of the meter bridge.
- B. To determine the temperature coefficient of resistance of a given wire.

### Materials and Apparatus

Meter bridge, three standard resistors (2  $\Omega$ , 5  $\Omega$ , 10  $\Omega$ ), galvanometer, connecting wires, sliding key, Leclanche cell, resistance box, two keys, Bunsen burner, thermometer, stirrer, stands, wire wound on a glass test tube, and a beaker of ice.

### Theory

#### Part A:



**Fig. 1:** The experimental circuit to determine end corrections of the meter bridge  
 Where,  $R'$ : high resistance,  $P, Q$ : standard resistors,  $K_1, K_2$ : plug keys,  
 $G$ : galvanometer,  $l$ : balance length,  $E$ : Leclanche cell,  $e_1$ : end correction at A,  
 $e_2$ : end correction at B

At the balance point in Fig.1;

$$\frac{P}{Q} = \frac{(l_1 + e_1)}{[(100 - l_1) + e_2]}$$

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$$\frac{P}{P+Q} = \frac{(l_1 + e_1)}{[100 + e_1 + e_2]} \quad (1)$$

Interchange  $P$  and  $Q$  at the balance point;

$$\frac{Q}{P} = \frac{(l_2 + e_1)}{[(100 - l_2) + e_2]}$$

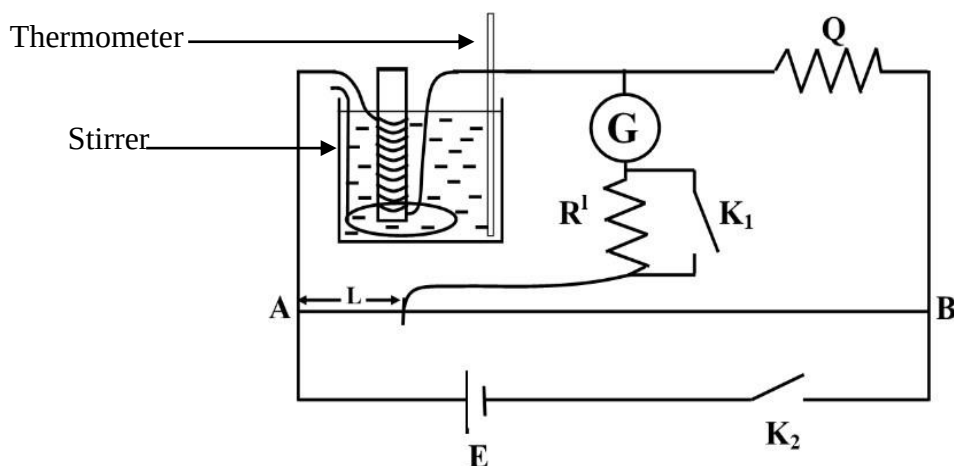
$$\frac{Q}{P+Q} = \frac{(l_2 + e_1)}{[100 + e_1 + e_2]} \quad (2)$$

From equations (1) and (2);

$$e_1 = \frac{(Ql_1 - Pl_2)}{[P - Q]} \quad (3)$$

$$e_2 = \left[ \frac{Q}{P} (l_1 + e_1) \right] - (100 - l_1) \quad (4)$$

### Part B



**Fig. 2:** The experimental circuit to determine the temperature coefficient of resistance of the given wire

At the balance point in Fig. 2;

$$\frac{R_T}{L + e_1} = \frac{Q}{(100 - L + e_2)} \quad (5)$$

The resistance of a given wire at temperature  $T$  given by,

$$R_T = R_o(1 + \alpha T + \beta T^2 + \gamma T^3 + \dots)$$

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where,

$R_T$ : The resistance of a given wire at temperature  $T$

$R_0$ : resistance at  $0^\circ\text{C}$

$\alpha$ : First-order Temperature Coefficient of Resistance

$\beta$ : Second-order Temperature Coefficient of Resistance

$\gamma$ : Third-order Temperature Coefficient of Resistance

Since  $\beta$  and  $\gamma$  are comparatively low than  $\alpha$ ,

$$R_T = R_0 (1 + \alpha T)$$

By arranging the above equation;

$$R_T = (R_0 \alpha) T + R_0 \quad (6)$$

### Procedure

#### **Part A:**

a) Arrange the apparatus as shown in Fig. 1.

**Q.** Why it is important to use high resistance for  $R'$  ( $500\ \Omega$ )?

**Q.** How can you find a rough balance point? Next, how do you measure the exact balance point?

**Q.** Can you find  $e_1$  and  $e_2$  without interchanging  $P$  and  $Q$ ?

#### **Part B:**

a) Arrange the apparatus as shown in Fig. 2 using the ice beaker when  $Q = 10\ \Omega$ .

b) Obtain a set of values for balance length  $L$  by varying the temperature of the given wire.

**Q.** Explain how you calculate  $R_T$  values.

**Q.** Explain how you obtain the  $\alpha$  from the graph?